

INSTITUTE OF ACTUARIES OF INDIA
EXAMINATIONS

May 2026

SP1- Health and Care

Time allowed: 3 Hours 15 Minutes

Total Marks: 100

Q.1) A mid-sized health insurer offers family floater indemnity health products (₹3–50 lakh sum insured, cashless at 1,200 network hospitals, 3-year pre-existing disease waiting period). Loss ratio deteriorated from 94% to 118% over last three years due to:

- High-value cancer/cardiac claims (ages 45–60 years);
- Suspected fraudulent short-stay claims from network hospitals in one region;
- Underwriting lapses from volume-incentivised telesales agents.

i) Identify distinct sources of risk in the health insurance portfolio and classify into risk type, also briefly justify each risk specific to the context.

[5]

ii) With reference to the underwriting cycle:

a) Explain how the telesales agents' volume-based incentive structure contributes to adverse selection in the floater portfolio. **[3]**

b) Recommend four specific, actionable underwriting controls with rationale to restore risk selection discipline. **[4]**

iii) The suspected fraud cluster from network hospitals represents a material operational and insurance risk.

- Assess the indicators of provider fraud present in this scenario, and
- Describe a structured fraud detection and investigation framework that insurer's third-party administrators and claims team may implement.

[7]

[19 marks]

Q.2) A standalone health insurer, plans to launch a new retail indemnity health product with comprehensive in-patient cover (including day-care, domiciliary, home-care and AYUSH), no room-rent caps, entry up to age 75 years, pre-existing waiting of 36 months, and no-claim cumulative bonus.

The insurer plans aggressive pricing and a quota-share plus excess-of-loss reinsurance programme for its health portfolio.

i) List the key regulatory requirements to be complied and constraints relevant to:

- Design, pricing and governance of the product; and
- The health reinsurance programme, explaining briefly how they support protection of policyholders interests.

[7]

ii) Explain the factors that should be considered when determining the pricing and reserving approach for the product, considering the applicable regulatory framework and the proposed reinsurance structure.

[8]

[15 marks]

Q.3) A stand-alone health insurer manages a large group health scheme covering 80,000 lives for a textile conglomerate across multiple locations. Due to competitive pricing pressure, the scheme has renewed at flat rates despite experiencing loss ratios exceeding 140% over three years. The scheme is due for renewal in 60 days, with the corporate client indicating they will switch insurers if premiums increase by more than 15%.

The Board seeks a strategic risk review for the health portfolio, which has also shown overall deterioration in claims experience.

- i) Critically evaluate the options, for the group scheme renewal, available to the insurer in managing the financial and reputational risks, and make a justified recommendation. [8]
- ii) Propose a risk management framework for the health portfolio, including risk appetite metrics (quantitative/qualitative), its rationale and triggers for management action for insurer's health portfolio. [10]
- iii) The insurer is considering purchasing proportional reinsurance to manage volatility in its health portfolio. Discuss the considerations the actuarial team should evaluate before recommending a reinsurance structure. [5]

[23 marks]

Q.4) A health insurance company that has recently launched an individual Critical Illness (CI) product with tiered benefits. On diagnosis of an early-stage specified condition, 25% of the sum assured is paid; on subsequent progression to an advanced stage, the remaining 75% is paid. The policy ceases on full payment, death, or end of the policy term. This is company's first tiered CI product and it has limited relevant internal data.

- i) Describe the main features of a multi-state model appropriate for this product, clearly identifying the states and transitions between them. [4]
- ii) The actuarial team in the company is deciding between a stochastic and a deterministic approach for modelling this product. Describe the key differences between the two approaches and state, with reasons, the suitability for the company at this stage. [4]
- iii) The actuarial team builds a deterministic cashflow model and proposes carrying out sensitivity analysis before finalising premium rates. Describe about sensitivity analysis should be carried out and highlighting its importance for this product given company's circumstances context. [4]

After the first year of selling the tiered CI product, Company's actuarial team carries out an analysis of surplus. The following table sets out the actual versus expected experience for the first year:

Experience	Expected	Actual
Early-stage CI inception rate (per 1,000 lives)	8	11.5
Progression rate: early-stage → advanced-stage	20%	14%
Annual lapse rate	8%	13%
Investment return	6%	7%
Expense per policy (₹)	1,200	1,470

iv)

- a) Briefly explain about analysis of surplus, and describe how it is structured. [2]
 - b) For each item in the table above, state with reasoning whether the variance from expected is favourable or adverse to the Company. [5]
 - c) Based on the overall experience in the table, discuss the actions that Company's management and actuary should consider. [3]
- [22 marks]**

Q.5)

- i) Describe the main features of a top-up health insurance product. [3]
- ii) Discuss the factors that should be considered when designing the product features of a top-up health insurance product. [5]
- iii) Assess the relative merits of a top-up health insurance product compared to a comprehensive indemnity health insurance product from the perspective of a policyholder. [2]

[10 marks]

Q.6) An insurer offering individual indemnity-based product is considering adding a no-claim discount (NCD) feature to its product. Under the proposed NCD structure, policyholders who make no claim in a policy year receive a discount on their renewal premium.

- i) Describe the equation of value approach for pricing a one-year individual indemnity-based product, and state one limitation of this approach in this context. [3]
- ii) Discuss the advantages and disadvantages of introducing the NCD feature from the insurer's perspective. [5]
- iii) Discuss how the NCD feature should affect the insurer's approach to premium rating, and the risks that arise if premium rates are not adjusted appropriately. [3]

[11 marks]
